

BABU BANARASI DAS UNIVERSITY LUCKNOW



SCHOOL OF COMPUTER APPLICATIONS

Project Report File Customer Churn Analysis and Prediction Using Data Analytics and Machine Learning

Submitted to:

**Mr. Harendra Singh Sir
IBM**

Submitted by:

**Satwik Srivastava
Roll No. 05 / 1250258395
BCA DS - 26
BCA (DS & AI)
(2026 - 2027)
25/09/2026**

ACKNOWLEDGEMENT

I sincerely express my gratitude to my teacher, Mr. Harendra Singh, for providing me with the opportunity to work on the project titled “Customer Churn Analysis and Prediction Using Data Analytics and Machine Learning.” His academic guidance and direction helped me understand how a practical data analytics problem can be approached systematically, beginning with dataset exploration and continuing through statistical analysis, exploratory data analysis, machine learning, model evaluation, and interpretation.

This project provided an opportunity to apply concepts studied as part of the BCA (Data Science & AI) curriculum to a customer analytics problem. While working on the project, I developed practical understanding of Python-based data analysis, Pandas and NumPy, visualization, preprocessing of numerical and categorical variables, feature transformation, classification algorithms, cross-validation, hyperparameter tuning, and model interpretation.

I am thankful for the support and academic environment that made it possible to complete the work. The project also helped me understand that machine learning is not limited to training a model; reliable results depend on data quality, appropriate preprocessing, meaningful exploratory analysis, suitable evaluation metrics, and careful interpretation of the final results.

Finally, I would like to acknowledge the value of hands-on experimentation during the development of this project. The complete workflow—from loading the Excel dataset to generating churn predictions—helped strengthen my understanding of how data analytics and machine learning can be integrated into one end-to-end solution.

INTRODUCTION

Customer churn refers to the situation in which an existing customer stops using a company's products or services. For businesses that depend on recurring customers, understanding churn is important because customer loss affects revenue, customer relationships, service planning, and long-term business performance. A structured analysis of customer information can reveal patterns associated with churn and can help organizations understand which customer characteristics and service-related factors deserve further attention.

The present project, "Customer Churn Analysis and Prediction Using Data Analytics and Machine Learning," combines descriptive data analytics with supervised machine learning. The analytical component focuses on understanding the customer dataset through statistical summaries and exploratory visualizations. The machine learning component uses the same prepared customer information to build classification models capable of predicting whether a customer is likely to churn.

The project uses an Excel dataset containing 20,000 customer records and 31 variables. The dataset includes demographic information, customer tenure, service subscriptions, contract details, billing information, usage-related variables, support activity, satisfaction, city tier, referral information, and the final Churn status. This combination makes it possible to examine churn from multiple perspectives rather than relying on a single customer attribute.

The workflow begins with loading and inspecting the dataset, followed by statistical analysis and data quality checks. Exploratory data analysis is then performed using categorical and numerical variables. After this analytical stage, the Churn variable is encoded as the target, non-predictive CustomerID is removed, categorical variables are encoded, numerical variables are scaled, and the data is divided into training and testing sets. Multiple classification algorithms are then developed and compared.

The project finally applies cross-validation and hyperparameter tuning to the Random Forest model and evaluates the resulting model using accuracy, precision, recall, F1-score, ROC-AUC, a confusion matrix, a ROC curve, and a classification report. The complete workflow demonstrates how an analytical question can be transformed into a reproducible machine learning solution.

PROBLEM STATEMENT

Customer churn is a classification problem in which the objective is to identify whether an existing customer will remain with a service provider or leave. A customer dataset can contain many variables, and the relationship between these variables and churn may not be immediately visible. Therefore, simply looking at individual columns is not sufficient to understand the problem.

The problem addressed in this project is to analyze customer-level data and identify patterns related to churn while also developing machine learning models that can classify customers according to their churn status. The analysis needs to consider demographic characteristics, contract information, service usage, billing variables, tenure, customer satisfaction, and support activity.

Another important aspect of the problem is preparing heterogeneous customer data for machine learning. The dataset contains both numerical and categorical attributes. Numerical variables such as Age, Income, TenureMonths, MonthlyCharges, TotalCharges, AvgMonthlyUsageGB, NumSupportTickets, LatePayments, and SatisfactionScore require numerical processing, while categorical variables such as Gender, MaritalStatus, Education, Contract, InternetService, and PaymentMethod require encoding.

The project therefore addresses the complete analytical problem rather than only model training. It examines data quality, explores distributions and relationships, prepares the data through a preprocessing pipeline, trains several classification algorithms, compares their performance, performs stratified cross-validation, tunes a Random Forest model, and interprets important features.

The final output is intended to provide a clear analytical and predictive view of customer churn. The results are presented through tables, charts, evaluation metrics, feature importance, and customer-level churn predictions. This makes the project suitable as an integrated Data Analytics and Machine Learning study.

OBJECTIVES

The main objective of this project is to develop an end-to-end customer churn analysis and prediction workflow using data analytics and machine learning. The objectives are designed to cover both the exploratory and predictive parts of the project.

- To load and inspect the customer churn dataset from the raw_data Excel sheet.
- To understand the structure, dimensions, columns, data types, and sample observations of the dataset.
- To calculate descriptive statistics for numerical variables and summarize the overall dataset.
- To perform data quality analysis and identify missing values or other data-quality concerns.
- To analyze categorical and numerical variables using appropriate statistical summaries and visualizations.
- To study the distribution of churn and examine relationships between churn and customer characteristics.
- To prepare the dataset for machine learning by encoding the target and categorical features and scaling numerical features.
- To divide the dataset into training and testing subsets using stratified sampling.
- To develop Logistic Regression, Decision Tree, Random Forest, and Gradient Boosting classification models.
- To evaluate models using Accuracy, Precision, Recall, F1-Score, and ROC-AUC.
- To perform 5-fold stratified cross-validation and compare model stability.
- To tune the Random Forest model using GridSearchCV.
- To identify important features associated with the final tuned model.
- To generate customer-level churn predictions and probabilities.

- To summarize the major findings and demonstrate how analytics and machine learning can be combined in one project.

DATASET DESCRIPTION

The project uses an Excel-based customer dataset loaded from the sheet named “raw_data”. The dataset contains 20,000 customer records and 31 columns. Each row represents a customer and contains information describing demographic characteristics, services, billing, usage, support activity, satisfaction, and churn status.

<u>Dataset Attribute</u>	<u>Value</u>
Source format	Excel
Sheet	raw_data
Number of records	20,000
Number of columns	31
Target variable	Churn
Numerical columns	9 before preprocessing
Categorical columns	20 before preprocessing
Identifier	CustomerID

The CustomerID field provides a customer identifier and is not used as a predictive feature. The Churn field represents the target variable and contains the customer’s final churn status. The dataset includes both continuous numerical measures and categorical business attributes, making preprocessing an important part of the machine learning workflow.

Important variables include Age, SeniorCitizen, Income, TenureMonths, MonthlyCharges, TotalCharges, AvgMonthlyUsageGB, NumSupportTickets, LatePayments, SatisfactionScore, Contract, InternetService, PaymentMethod, and several service-related fields. Together, these variables allow the project to investigate whether customer behaviour, service configuration, financial characteristics, and customer experience are associated with churn.

The dataset contains 12,449 customers with Churn = No and 7,551 customers with Churn = Yes. Thus, the overall observed churn rate in the dataset is 37.76%. This target distribution is retained through stratified train-test splitting so that the training and testing datasets preserve the class proportions.

TOOLS AND TECHNOLOGIES

The project was implemented using Python and commonly used data analytics and machine learning libraries. Google Colab was used as the working environment for executing the notebook, inspecting outputs, generating visualizations, training models, and exporting the completed project.

<u>Tool / Library</u>	<u>Purpose in Project</u>
Python	Core programming language
Pandas	Data loading, manipulation and analysis
NumPy	Numerical computation
Matplotlib	Data visualization
Seaborn	Statistical visualization
Scikit-learn	Preprocessing, classification, evaluation, cross-validation and tuning
Statsmodels	Statistical/time-series related analytical support where required
Excel	Dataset storage and input format
Google Colab	Notebook execution environment

Pandas is used to read the Excel dataset and perform operations such as inspection, filtering, descriptive statistics, and feature preparation. NumPy supports numerical operations. Matplotlib and Seaborn are used to create charts for churn distribution, demographic analysis, contract analysis, charges, tenure, satisfaction, support tickets, payment methods, and correlations.

Scikit-learn provides the machine learning workflow. Its preprocessing components are used to scale numerical features and one-hot encode categorical features. Its classification algorithms are used to train the four

selected models. Evaluation utilities are used for classification metrics, confusion matrices, ROC curves, cross-validation, and GridSearchCV.

The use of these tools makes the project reproducible and provides a standard workflow for transforming raw customer information into analytical insights and machine learning predictions.

METHODOLOGY

The project follows a structured end-to-end methodology that connects Data Analytics with Machine Learning. The first stage is dataset loading and understanding. The Excel sheet `raw_data` is read into a Pandas DataFrame, after which the dimensions, columns, data types, sample records, and statistical properties are examined.

The second stage is data quality and exploratory analysis. Missing values are checked across all columns, and categorical and numerical variables are examined separately. Exploratory analysis uses charts and summaries to understand churn distribution and investigate variables such as age, gender, contract type, monthly charges, tenure, satisfaction score, support tickets, internet service, payment method, and correlations among numerical attributes.

The third stage is machine learning preparation. Churn is converted from No/Yes into 0/1. CustomerID is removed because it is an identifier rather than a meaningful predictive variable. Numerical and categorical features are identified separately. Numerical features are standardized using StandardScaler, while categorical variables are transformed using OneHotEncoder. A preprocessing pipeline combines these operations.

The fourth stage is model development. Four classification models are trained: Logistic Regression, Decision Tree, Random Forest, and Gradient Boosting. The models are evaluated on the test set using multiple metrics so that performance is not judged using accuracy alone.

The fifth stage is validation and optimization. Stratified 5-fold cross-validation is performed to examine model performance across multiple training-validation splits. Random Forest is then tuned using GridSearchCV over a defined parameter grid. The best parameters are selected based on cross-validated ROC-AUC, after which the tuned model is evaluated on the held-out test set.

The final stage is interpretation. Feature importance from the tuned Random Forest is examined, customer-level predictions and churn probabilities are generated, and final visual outputs include the confusion matrix, ROC curve, and classification report. The project concludes with findings based on the actual analytical and model results.

INITIAL DATASET OVERVIEW

The initial dataset overview establishes the basic structure of the customer data before statistical analysis or machine learning preparation. The Excel file is loaded into a Pandas DataFrame using the raw_data sheet. The first and last observations are inspected to confirm that records have been read correctly.

The dataset contains 20,000 rows and 31 columns. The columns represent a mixture of identifiers, demographic characteristics, customer service information, financial attributes, usage measures, support activity, and the target churn variable. The dataset therefore provides a broad customer-level view suitable for exploratory and predictive analysis.

<u>Initial Check</u>	<u>Result</u>
Rows	20,000
Columns	31
Target	Churn
Data types	22 object, 6 int64, 3 float64
Customer identifier	CustomerID
Target classes	No / Yes

The dataset structure is important because machine learning algorithms cannot directly use raw text categories in their original form. The later preparation stage therefore separates categorical and numerical variables and applies appropriate transformations.

The initial inspection also confirms that the target variable is available for supervised learning. The project does not treat CustomerID as a behavioural variable; it is removed before training. The remaining attributes provide the information from which models learn patterns associated with churn.

This initial overview creates the foundation for subsequent statistical analysis, data quality checks, EDA, preprocessing, and model development.

STATISTICAL ANALYSIS

Descriptive statistics are used to understand the central tendency, spread, and range of numerical variables. The project uses Pandas describe() to generate summary statistics and also examines categorical variables using the corresponding descriptive output.

<u>Variable</u>	<u>Mean</u>	<u>Minimum</u>	<u>Maximum</u>
Age	46.55	18	75
Income	98,652.51	18,002	179,978
TenureMonths	36.35	1	72
MonthlyCharges	79.91	20	140
TotalCharges	2,909.38	-11.10	10,059.35

The numerical summary shows that customer ages range from 18 to 75 years, while tenure ranges from 1 to 72 months. MonthlyCharges range from 20 to 140, and TotalCharges show a wide spread across customers. The average values provide a baseline understanding of the scale of each variable before preprocessing.

Statistical summaries are useful because they reveal the scale and distribution of variables and help determine whether standardization is required. They also support later interpretation of visualizations by providing numerical reference points.

The analysis is not limited to averages. Standard deviation, quartiles, minimum, and maximum values are also considered. These statistics provide information about variability and potential extreme values. Since machine learning preparation later standardizes numerical variables, the original scale differences are handled systematically.

DATA QUALITY ANALYSIS

Data quality is examined before exploratory analysis and machine learning. A missing-value check is performed across all 31 columns. The analysis shows zero missing values for every column in the dataset. Therefore, no missing-value imputation is required for the project.

<u>Quality Check</u>	<u>Result</u>
Total records	20,000
Total columns	31
Columns with missing values	0
Missing values requiring imputation	0
Target availability	Complete

The absence of missing values simplifies the preprocessing workflow because the project does not need to introduce mean, median, mode, or model-based imputation. This allows the analysis to proceed directly to variable exploration and machine learning transformation.

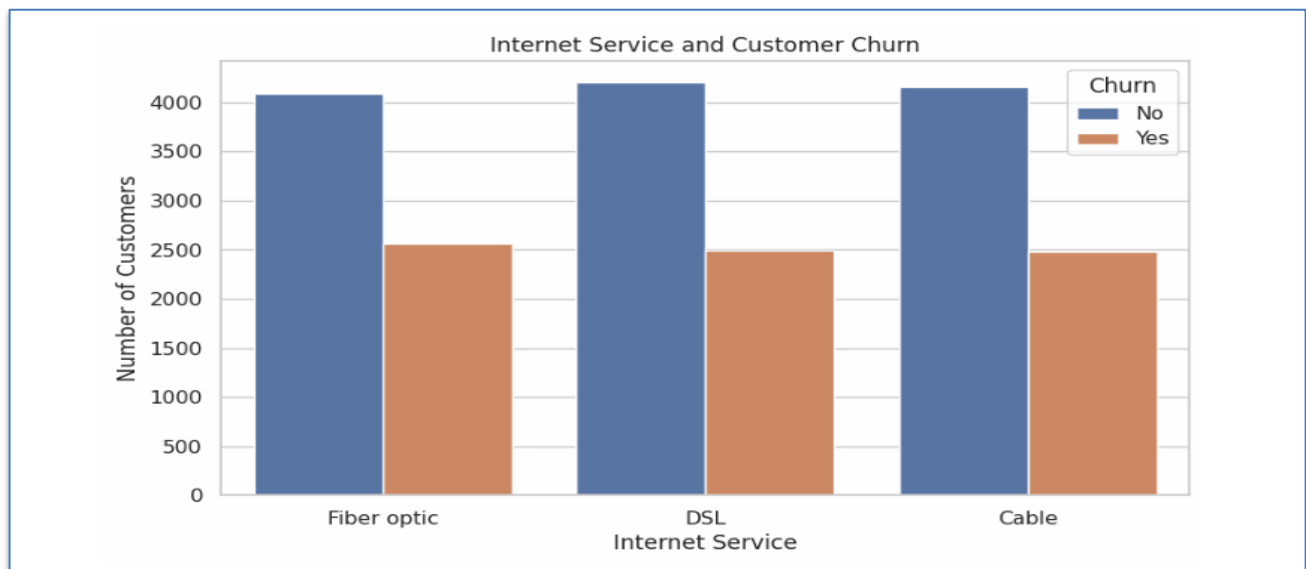
Data quality analysis is still important even when missing values are not present. It verifies that the dataset is structurally complete and prevents assumptions about data completeness from being made without inspection.

The project therefore records the data-quality result explicitly and continues with categorical analysis, numerical analysis, and exploratory visualizations. Any modelling decisions are made after this quality check rather than before it.

CATEGORICAL VARIABLE ANALYSIS

Categorical analysis examines customer attributes that represent groups or categories rather than continuous numerical measurements. The dataset contains variables such as Gender, MaritalStatus, Dependents, Education, EmploymentStatus, PhoneService, MultipleLines, InternetService, Contract, PaperlessBilling, PaymentMethod, CityTier, Referral, and Churn.

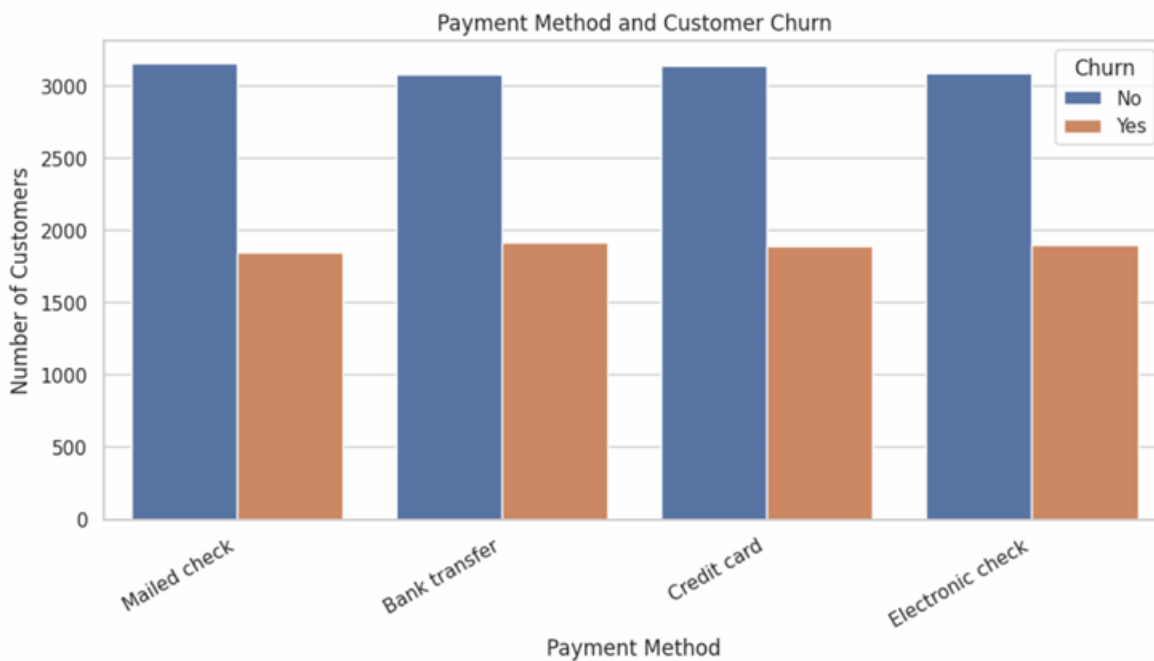
Frequency-based visualizations are used to understand category distributions and compare churn status within selected categories. Contract type is specifically analyzed because it represents the customer's contractual relationship with the service provider. Internet service and payment method are also examined to identify differences in churn distribution across service and billing categories.



Insight: The graph compares churn counts across DSL, Fiber optic, and Cable internet-service categories. It provides a service-level view of how churn is distributed across different internet-service groups.

The analysis of categorical variables provides a descriptive view of how customer groups are represented in the dataset. Rather than considering churn as an isolated outcome, the project examines it alongside customer attributes and service configurations.

These categorical variables are also important during machine learning preparation. Because classification algorithms require numerical input, categorical attributes are later transformed using one-hot encoding. This creates separate binary features for the relevant categories while preserving the information contained in the original variables.



Insight: The visualization compares churn counts across Credit card, Mailed check, Bank transfer, and Electronic check categories, providing a billing-related perspective on customer churn.

The categorical analysis therefore serves two purposes: it supports exploratory understanding of the customer population and prepares the reasoning for the feature transformation used during modelling.

NUMERICAL VARIABLE ANALYSIS

Numerical variable analysis focuses on customer attributes measured on numeric scales. The project examines Age, Income, TenureMonths, MonthlyCharges, TotalCharges, AvgMonthlyUsageGB, NumSupportTickets, LatePayments, and SatisfactionScore.

Visualizations such as histograms and box plots are used to examine distributions and compare numerical variables across churn status. Age provides demographic context, while tenure indicates the length of the customer relationship. MonthlyCharges and TotalCharges represent billing-related information. Support tickets and satisfaction score provide customer-experience measures.

Numerical analysis is particularly useful because many machine learning models can use these variables directly after appropriate scaling. The project applies StandardScaler during preprocessing so that variables measured on different scales can be transformed into comparable standardized values.

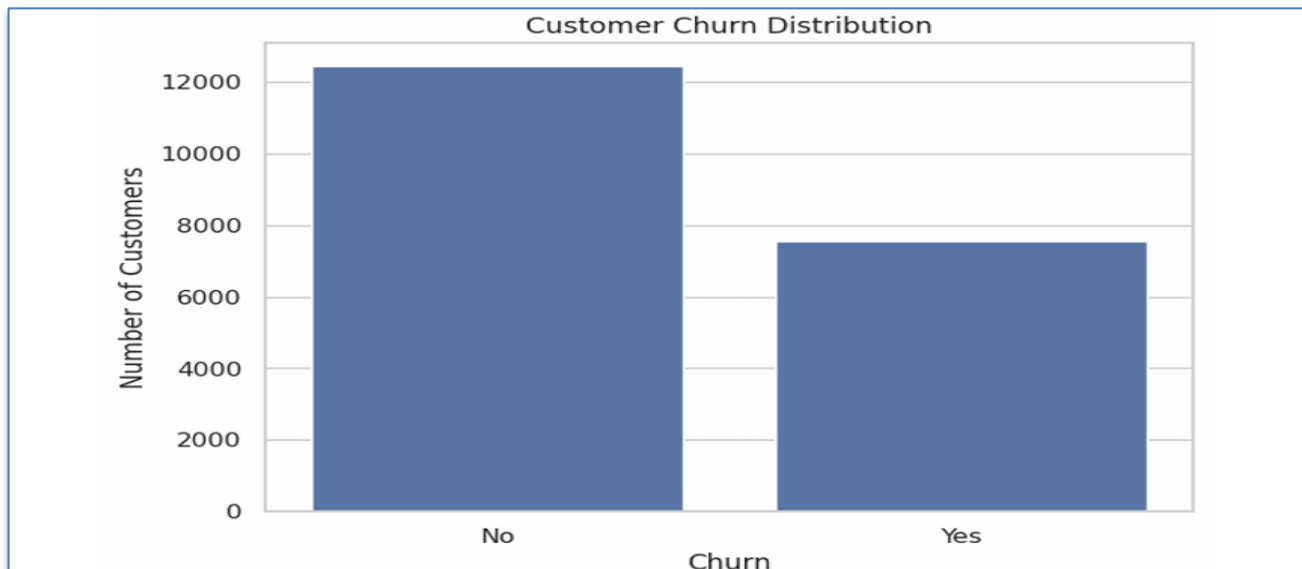
The numerical variables also contribute to the interpretation of the final model. Feature importance analysis later shows which transformed variables receive greater importance in the tuned Random Forest model. This connects the exploratory numerical analysis with the predictive stage.

Overall, numerical analysis helps establish the range, distribution, and practical meaning of the quantitative variables before they are passed into the machine learning pipeline.

EXPLORATORY DATA ANALYSIS (EDA)

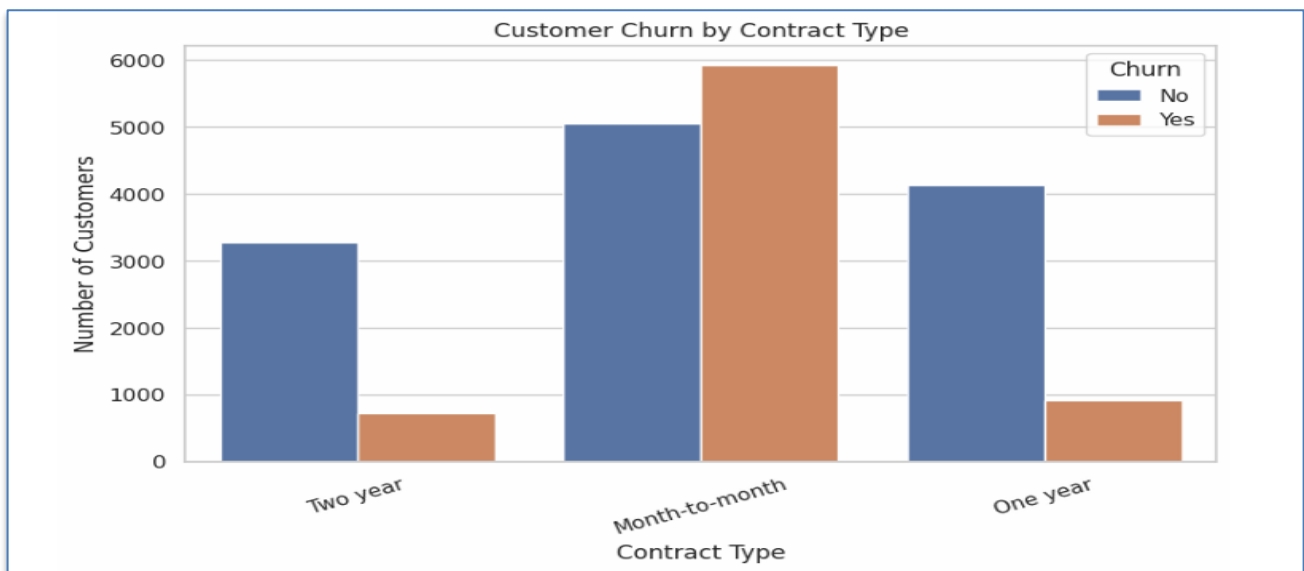
Exploratory Data Analysis is the central analytical stage of the project. The objective is to examine the distribution of churn and identify relationships between churn and customer characteristics. The project uses multiple visualizations rather than relying on a single chart, because churn can be associated with several dimensions simultaneously.

The churn distribution shows 12,449 customers with No churn and 7,551 customers with Yes churn, corresponding to an overall churn rate of 37.76%. This provides the baseline against which category-specific and numerical comparisons can be understood.

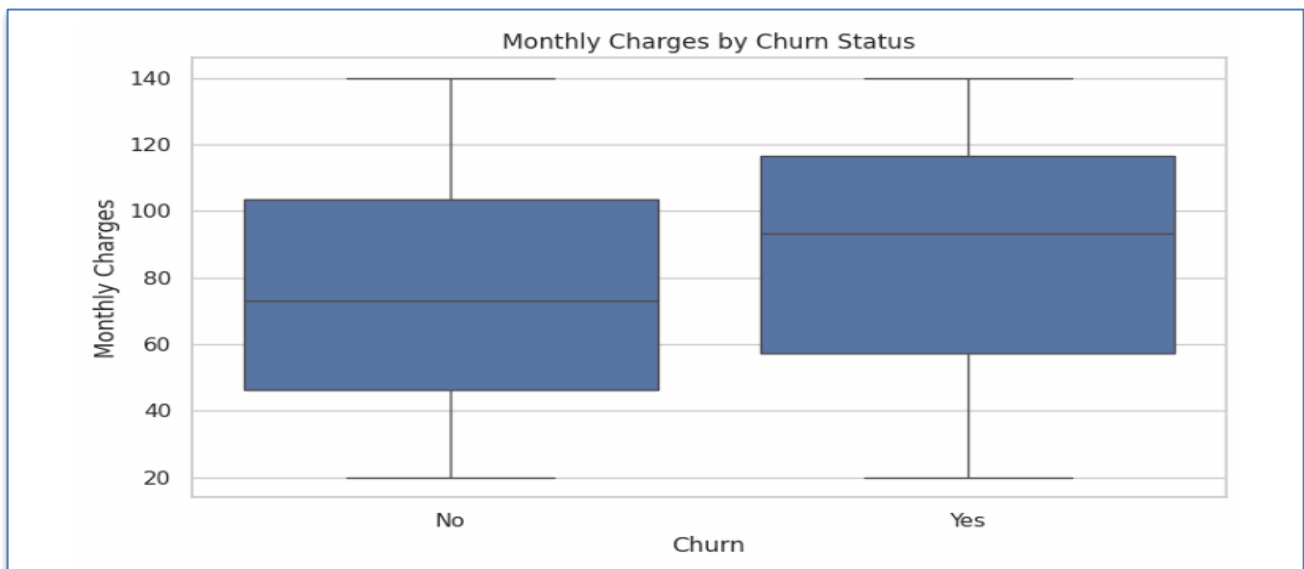


Insight: The dataset contains 12,449 customers who did not churn and 7,551 customers who churned. The overall churn rate is 37.76%, showing that churn represents a substantial portion of the customer base.

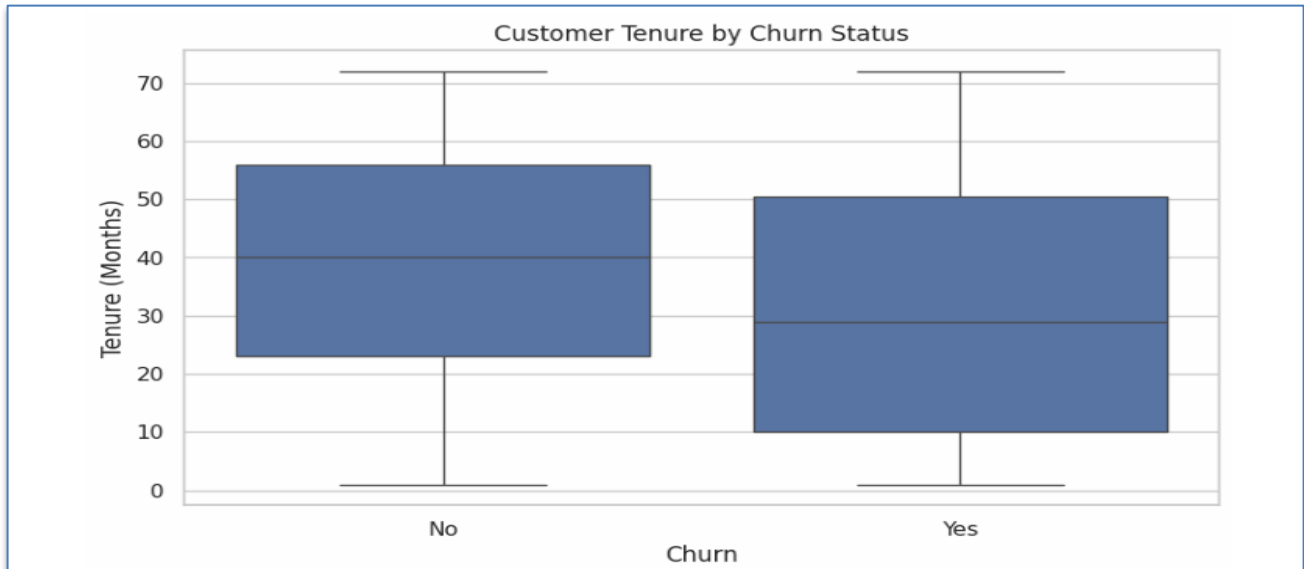
EDA covers customer age and gender, contract type, monthly charges, tenure, satisfaction score, support tickets, internet service, payment method, and numerical correlations. Contract type is examined using churn counts across contract categories. Monthly charges are compared across churn status. Tenure analysis investigates customer relationship length, while satisfaction and support-ticket analyses examine customer experience and support activity.



Insight: The graph compares churn across Month-to-month, One-year, and Two-year contracts. It provides a visual comparison of churn counts across different contractual arrangements and helps identify contract type as an important variable for further analysis.

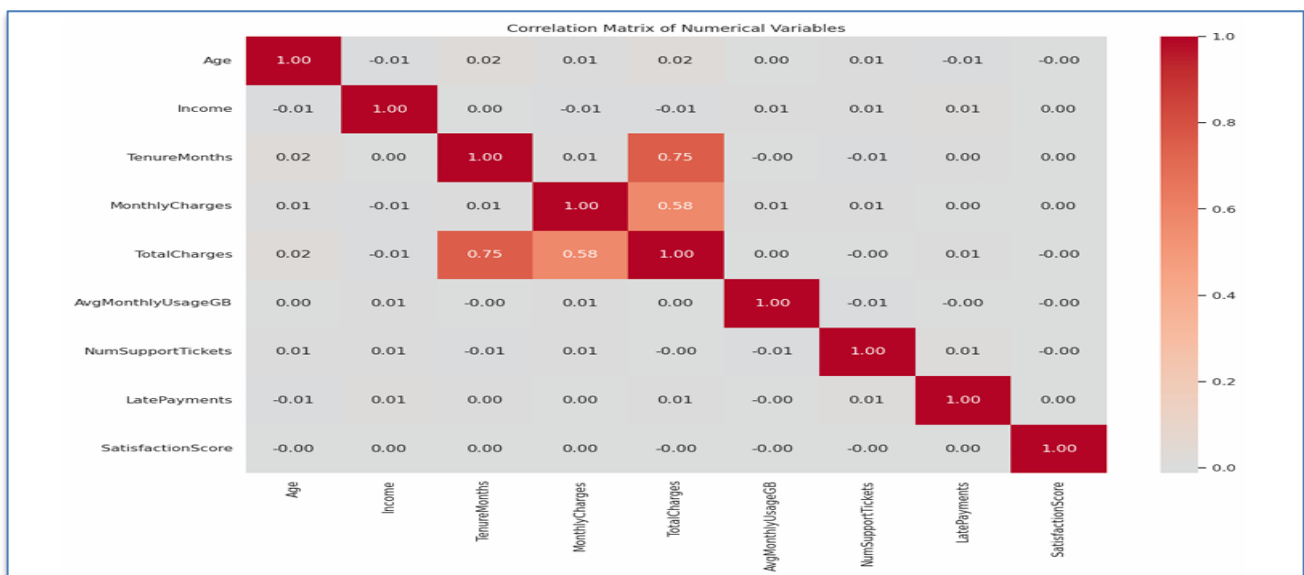


Insight: The box plot compares the distribution of monthly charges between churned and non-churned customers. It helps examine differences in charge levels and the spread of monthly charges across the two churn groups.



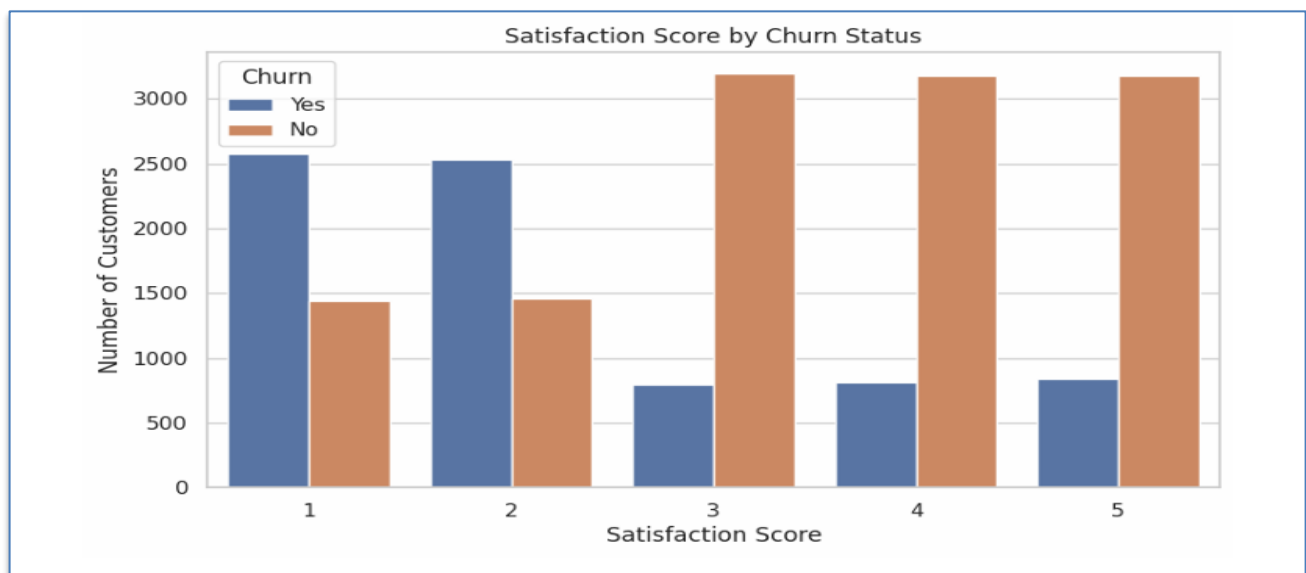
Insight: The box plot compares customer tenure between churn groups. Tenure is particularly relevant because it represents the length of the customer relationship and later appears among the important features in the tuned Random Forest model.

The correlation matrix provides an additional numerical perspective by showing pairwise relationships among numerical variables. This is useful for understanding whether certain measures move together and for identifying relationships that may be relevant during model interpretation.



Insight: The correlation heatmap provides a compact view of relationships among numerical customer attributes. It helps identify variables that move together and provides additional context for the later machine-learning analysis.

The EDA stage is not itself a prediction model. Its role is to understand the data, identify meaningful patterns, and provide context for later modelling decisions. The results guide the selection and interpretation of features while the machine learning models provide a quantitative predictive evaluation.



Insight: The graph compares satisfaction-score distributions across churn groups. Satisfaction is particularly important in this project because SatisfactionScore has the highest feature importance (0.255602) in the tuned Random Forest.

MACHINE LEARNING PREPARATION

After completing EDA, the dataset is prepared for supervised classification. The Churn column is selected as the target variable because the purpose of the machine learning component is to predict whether a customer will churn. Churn values are encoded as 0 for No and 1 for Yes.

CustomerID is removed because it is an identifier and does not represent a meaningful customer behaviour or characteristic for prediction. The remaining variables are separated into numerical and categorical groups.

<u>Preparation Step</u>	<u>Implementation</u>
Target encoding	No = 0, Yes = 1
Identifier handling	CustomerID removed
Numerical processing	StandardScaler
Categorical processing	OneHotEncoder
Train-test split	80% training / 20% testing
Training records	16,000
Testing records	4,000
Sampling strategy	Stratified

A preprocessing pipeline is used to apply transformations consistently. Numerical features are standardized, while categorical features are one-hot encoded. The use of a pipeline reduces the risk of applying inconsistent preprocessing between training and testing data.

The 80/20 stratified split creates 16,000 training records and 4,000 testing records while preserving the target-class distribution. The test set is kept separate for final evaluation so that model performance is measured on unseen customer records.

MACHINE LEARNING MODEL DEVELOPMENT

Four classification algorithms are developed to provide a comparative machine learning analysis. Logistic Regression is included as a linear baseline model. Decision Tree provides an interpretable rule-based classification approach. Random Forest combines multiple decision trees, while Gradient Boosting builds an ensemble sequentially to improve predictive performance.

<u>Model</u>	<u>Configuration / Role</u>
Logistic Regression	Baseline linear classifier
Decision Tree	max_depth=6, min_samples_split=10, min_samples_leaf=5
Random Forest	200 estimators, max_depth=10, min_samples_split=10, min_samples_leaf=4
Gradient Boosting	150 estimators, learning_rate=0.05, max_depth=3

Each model is trained using the same prepared feature representation so that the comparison focuses on algorithmic performance rather than different preprocessing. Predictions are generated on the held-out test set and evaluated using Accuracy, Precision, Recall, F1-Score, and ROC-AUC.

The model development stage demonstrates why multiple algorithms are useful. A single model may perform differently depending on the structure of the data. Comparing several approaches provides a broader understanding of predictive performance and supports later validation and tuning.

The models are also evaluated beyond accuracy. Precision indicates the proportion of predicted churn cases that are actually churn cases, recall measures how many actual churn cases are detected, F1-score balances precision and recall, and ROC-AUC measures ranking performance across classification thresholds.

MODEL EVALUATION

The initial model comparison is performed on the held-out test set. The results show differences between the four classifiers and provide a baseline before Random Forest tuning.

<u>Model</u>	<u>Accuracy</u>	<u>Precision</u>	<u>Recall</u>	<u>F1-Score</u>	<u>ROC-AUC</u>
Logistic Regression	0.8285	0.7865	0.7490	0.7673	0.8837
Decision Tree	0.9540	0.9955	0.8821	0.9354	0.9410
Random Forest	0.9322	0.9725	0.8444	0.9039	0.9301
Gradient Boosting	0.9520	0.9889	0.8828	0.9328	0.9331

The table demonstrates that model performance depends on the selected evaluation metric. Accuracy, precision, recall, F1-score, and ROC-AUC provide different perspectives on classification quality. The project therefore reports all of these measures instead of using a single score.

The Decision Tree records an ROC-AUC of 0.9410 in the original model comparison, while Gradient Boosting records 0.9331 and Random Forest records 0.9301. Logistic Regression records 0.8837. These values establish the baseline against which the later tuned Random Forest model is evaluated.

The evaluation stage also includes confusion matrices, ROC curves, precision-recall curves, and classification reports so that numerical metrics can be interpreted alongside class-level prediction results.

CROSS-VALIDATION

Cross-validation is used to examine whether model performance remains reasonably stable across multiple train-validation partitions. The project uses stratified 5-fold cross-validation, which maintains the class distribution while creating five different validation folds.

<u>Model</u>	<u>Mean ROC-AUC</u>	<u>Standard Deviation</u>
Logistic Regression	0.8791	0.0059
Decision Tree	0.9354	0.0060
Random Forest	0.9329	0.0054
Gradient Boosting	0.9347	0.0045

The mean ROC-AUC represents the average validation performance across the five folds, while the standard deviation indicates the variation among folds. Reporting both values gives more information than reporting a single validation score.

Gradient Boosting has a mean ROC-AUC of 0.9347 with a standard deviation of 0.0045. Decision Tree has a mean ROC-AUC of 0.9354 with a standard deviation of 0.0060, while Random Forest has 0.9329 ± 0.0054 . Logistic Regression records 0.8791 ± 0.0059 .

Cross-validation provides a more robust assessment of generalization because the model is evaluated repeatedly on different validation partitions. It is especially useful before hyperparameter tuning because it helps determine whether a model's performance is consistent rather than dependent on a single split.

HYPERPARAMETER TUNING

Hyperparameter tuning is performed for the Random Forest model using GridSearchCV. The objective is to systematically evaluate combinations of selected Random Forest parameters and identify a configuration that performs well under cross-validation.

<u>Parameter</u>	<u>Values Tested</u>
n_estimators	100, 200
max_depth	8, 12
min_samples_split	5, 10
min_samples_leaf	2, 4
Total combinations	16
Cross-validation fits	48

The best parameter combination obtained through the grid search is n_estimators=100, max_depth=12, min_samples_split=10, and min_samples_leaf=2. The best cross-validated ROC-AUC is 0.9346.

The tuning process demonstrates the difference between model development and model optimization. The original Random Forest configuration provides a baseline, while GridSearchCV evaluates systematically selected alternatives. The held-out test set remains separate and is used after tuning to measure final generalization performance.

Hyperparameter tuning is therefore treated as a controlled optimization stage rather than manual trial and error. The selected parameters are then used to train the final tuned Random Forest model for subsequent evaluation and interpretation.

FINAL MODEL COMPARISON

The final comparison includes the original four models and the tuned Random Forest model. This provides a consolidated view of the project's predictive results and shows how the tuned model compares with the original Random Forest and the other classifiers.

<u>Model</u>	<u>Accuracy</u>	<u>Precision</u>	<u>Recall</u>	<u>F1</u>	<u>ROC-AUC</u>
Logistic Regression	0.8285	0.7865	0.7490	0.7673	0.8837
Decision Tree	0.9540	0.9955	0.8821	0.9354	0.9410
Random Forest	0.9322	0.9725	0.8444	0.9039	0.9301
Gradient Boosting	0.9520	0.9889	0.8828	0.9328	0.9331
Tuned Random Forest	0.9418	0.9768	0.8662	0.9182	0.9317

	Model	Accuracy	Precision	Recall	F1-Score	ROC-AUC
0	Logistic Regression	0.8285	0.7865	0.7490	0.7673	0.8837
1	Decision Tree	0.9540	0.9955	0.8821	0.9354	0.9410
2	Random Forest	0.9322	0.9725	0.8444	0.9039	0.9301
3	Gradient Boosting	0.9520	0.9889	0.8828	0.9328	0.9331
4	Tuned Random Forest	0.9418	0.9768	0.8662	0.9182	0.9317

Insight: The comparison shows different performance levels across Logistic Regression, Decision Tree, Random Forest, and Gradient Boosting. The

original Decision Tree records the highest reported ROC-AUC of 0.9410, while Gradient Boosting records 0.9331, Random Forest 0.9301, and Logistic Regression 0.8837.

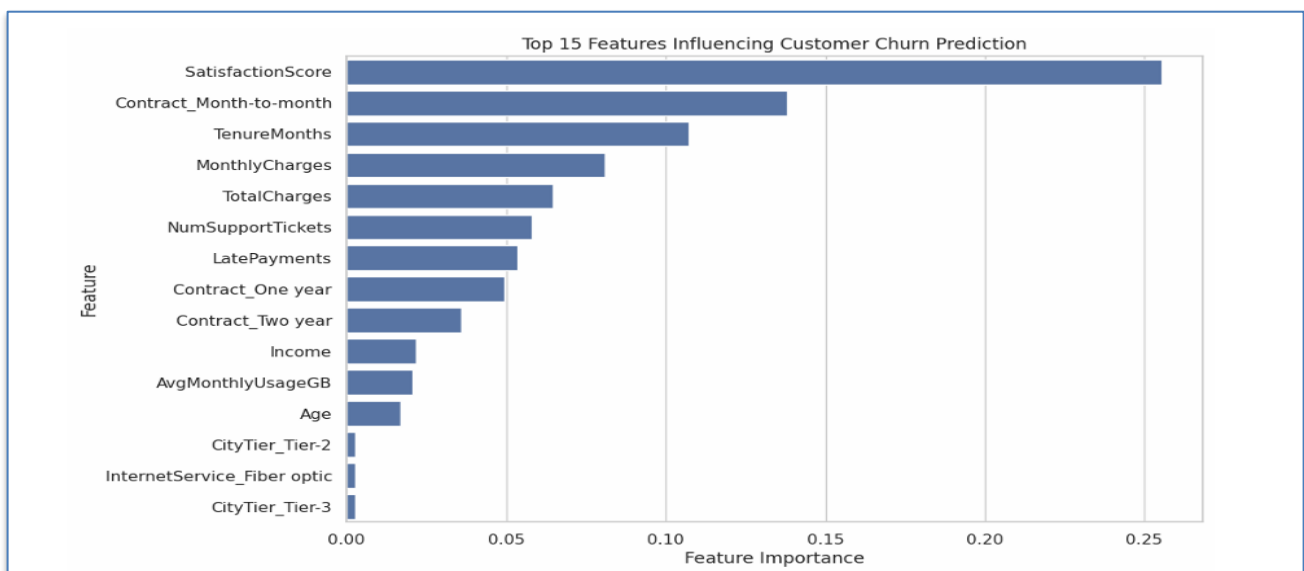
The tuned Random Forest achieves 0.9418 accuracy, 0.9768 precision, 0.8662 recall, 0.9182 F1-score, and 0.9317 ROC-AUC on the test set. Its cross-validated best ROC-AUC during tuning is 0.9346.

The comparison illustrates why multiple evaluation metrics are useful. The Decision Tree has an original test ROC-AUC of 0.9410, while the tuned Random Forest provides a different balance across accuracy, precision, recall, F1-score, and ROC-AUC. The report therefore records the numerical results without reducing model comparison to a single metric.

FEATURE IMPORTANCE

Feature importance is examined using the tuned Random Forest model to understand which transformed variables contribute most strongly to the model's predictive decisions. This provides an interpretation layer after model training and connects the machine learning output with the earlier exploratory analysis.

Rank	Feature	Importance
1	SatisfactionScore	0.255602
2	Contract_Month-to-month	0.138028
3	TenureMonths	0.107317
4	MonthlyCharges	0.080882
5	TotalCharges	0.064521
6	NumSupportTickets	0.057991
7	LatePayments	0.053526
8	Contract_One year	0.049375
9	Contract_Two year	0.035771
10	Income	0.021676



Insight: SatisfactionScore is the most important feature with importance 0.255602, followed by Contract_Month-to-month (0.138028), TenureMonths (0.107317), and MonthlyCharges (0.080882).

The highest reported feature importance is SatisfactionScore at 0.255602. Contract_Month-to-month has an importance of 0.138028, followed by TenureMonths at 0.107317 and MonthlyCharges at 0.080882. TotalCharges, NumSupportTickets, and LatePayments also appear among the leading features.

Feature importance should be interpreted as model-specific importance rather than proof of causation. The values indicate the contribution of transformed features to the tuned Random Forest's predictive process. They do not by themselves establish that changing a feature would directly cause a customer to churn or remain.

CUSTOMER CHURN PREDICTION

The trained tuned Random Forest model is used to generate customer-level churn predictions and churn probabilities for records in the test dataset. The prediction output combines the original test features with the actual churn label, predicted churn label, and predicted churn probability.

The predicted churn label is represented as No or Yes after converting the model's binary output back into the project's original terminology. The churn probability provides a continuous measure associated with the model's classification and can be used to inspect how strongly the model assigns a customer to the churn class.

<u>Prediction Output</u>	<u>Description</u>
Actual_Churn	Observed churn status in the test data
Predicted_Churn	Model-predicted churn status
Churn_Probability	Predicted probability associated with churn

Generating customer-level predictions is the practical application stage of the project. The model is no longer being evaluated only through aggregate metrics; it is being used to classify individual customer records.

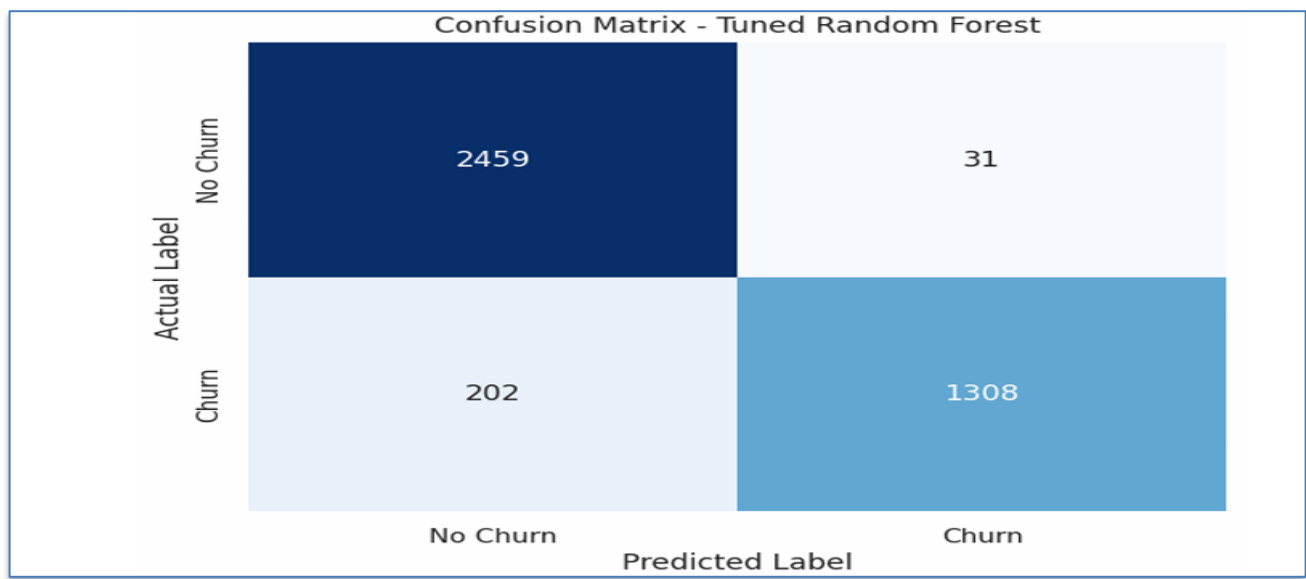
These predictions should be interpreted as model outputs rather than guaranteed future outcomes. The predictive system learns patterns from the available dataset and estimates churn status based on those patterns. The evaluation metrics reported in the project provide the context required to understand the model's predictive performance.

FINAL CONFUSION MATRIX

The final confusion matrix evaluates the tuned Random Forest by comparing actual churn status with predicted churn status on the test set. A confusion matrix provides a class-level view of correct and incorrect predictions and complements the aggregate metrics reported earlier.

The test dataset contains 4,000 customer records. The final classification report shows 2,490 No Churn cases and 1,510 Churn cases in the test set. The tuned model records a recall of 0.8662 for the churn class and a precision of 0.9768.

<u>Measure</u>	<u>Value</u>
Test records	4,000
No Churn support	2,490
Churn support	1,510
Churn precision	0.9768
Churn recall	0.8662
Churn F1-score	0.9182



Insight: The confusion matrix shows how the tuned Random Forest classified the 4,000 test records across No Churn and Churn classes. The churn class has 0.9768 precision, 0.8662 recall, and 0.9182 F1-score.

The confusion matrix is useful because accuracy alone does not show how errors are distributed between churn and non-churn classes. In a churn problem, identifying actual churn cases is particularly important for understanding recall, while precision indicates how reliable the predicted churn group is.

The project therefore presents the confusion matrix together with precision, recall, F1-score, and ROC-AUC rather than interpreting the matrix in isolation.

FINAL ROC CURVE

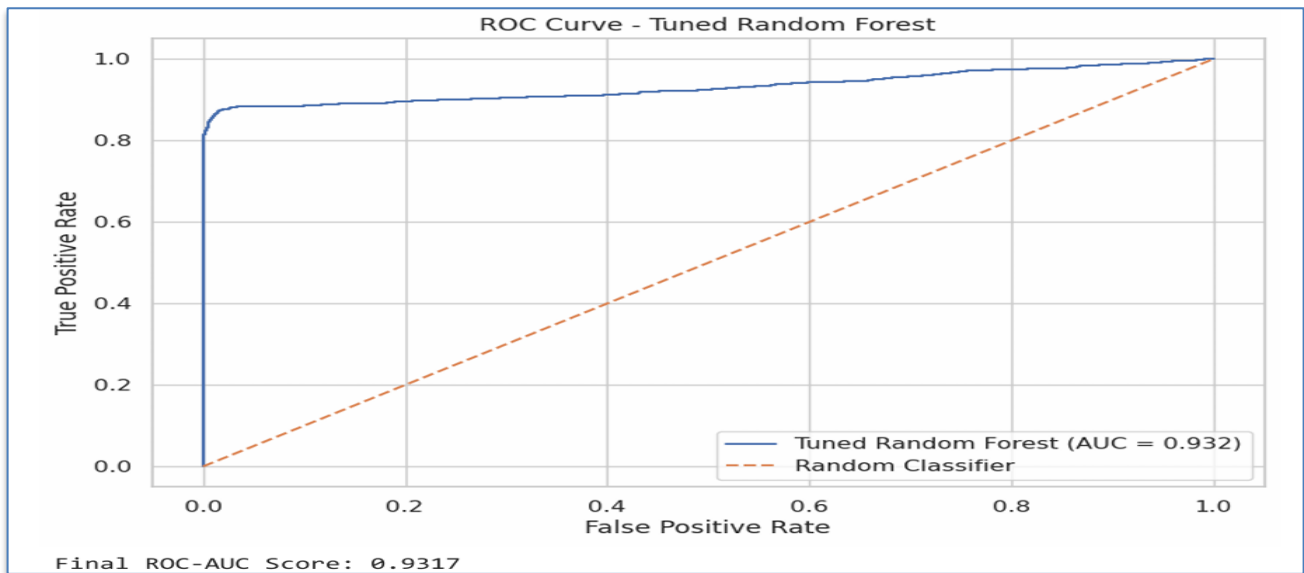
The Receiver Operating Characteristic (ROC) curve is used to visualize the classification performance of the tuned Random Forest across different decision thresholds. The curve compares the true positive rate with the false positive rate as the classification threshold changes.

The final tuned Random Forest records a test ROC-AUC of 0.9317. ROC-AUC summarizes the model's ability to distinguish between the two churn classes across classification thresholds.

<u>Final Model Metric</u>	<u>Value</u>
Model	Tuned Random Forest
Test Accuracy	0.9418
Test Precision	0.9768
Test Recall	0.8662
Test F1-Score	0.9182
Test ROC-AUC	0.9317

The ROC curve is useful because it evaluates ranking behaviour across multiple thresholds rather than using only the default classification threshold. The AUC value is therefore reported as an additional performance measure alongside the classification metrics.

The ROC analysis forms part of the final evaluation stage and is consistent with the project's approach of using multiple complementary measures to describe model performance.



Insight: The final tuned Random Forest achieves a ROC-AUC of 0.9317, indicating its classification performance across different decision thresholds.

FINAL CLASSIFICATION REPORT

The classification report provides class-specific precision, recall, and F1-score for the tuned Random Forest. It also reports macro and weighted averages, giving a consolidated summary of performance across the two classes.

<u>Class / Measure</u>	<u>Precision</u>	<u>Recall</u>	<u>F1-Score</u>	<u>Support</u>
No Churn	0.9241	0.9876	0.9548	2490
Churn	0.9768	0.8662	0.9182	1510
Macro Avg	0.9505	0.9269	0.9365	4000
Weighted Avg	0.9440	0.9417	0.9410	4000

For the Churn class, precision is 0.9768, recall is 0.8662, and F1-score is 0.9182. For the No Churn class, precision is 0.9241, recall is 0.9876, and F1-score is 0.9548. The reported accuracy is approximately 0.9417 in the classification report output, while the model comparison table records 0.9418 due to rounding/output presentation.

The classification report provides a more detailed view than a single accuracy value because it shows how the model performs separately for each class. This is particularly relevant in churn analysis because errors involving churn and non-churn customers have different interpretations.

FINAL PROJECT FINDINGS

The completed analysis provides both descriptive and predictive findings about the customer churn dataset. The dataset contains 20,000 customer records, of which 7,551 are labelled as churned and 12,449 as not churned. The resulting overall churn rate is 37.76%. No missing values are identified in the data-quality analysis.

The exploratory analysis examines churn in relation to demographic variables, contracts, charges, tenure, satisfaction, support activity, internet service, payment methods, and numerical relationships. This establishes the analytical context before machine learning is applied.

Four classification models are developed and compared. The original test results show that Logistic Regression records 0.8837 ROC-AUC, Decision Tree 0.9410, Random Forest 0.9301, and Gradient Boosting 0.9331. Five-fold cross-validation produces mean ROC-AUC values of 0.8791, 0.9354, 0.9329, and 0.9347 respectively.

Random Forest is then tuned using GridSearchCV. The selected parameters are `n_estimators=100`, `max_depth=12`, `min_samples_split=10`, and `min_samples_leaf=2`, with a best cross-validated ROC-AUC of 0.9346. On the test set, the tuned model records 0.9418 accuracy, 0.9768 precision, 0.8662 recall, 0.9182 F1-score, and 0.9317 ROC-AUC.

Feature importance analysis identifies SatisfactionScore, Contract_Month-to-month, TenureMonths, MonthlyCharges, TotalCharges, NumSupportTickets, and LatePayments among the leading transformed features. These findings connect the predictive model with the customer attributes examined during EDA.

CONCLUSION

This project demonstrates an integrated approach to Customer Churn Analysis and Prediction using Data Analytics and Machine Learning. The workflow begins with a structured Excel dataset containing 20,000 customer records and 31 variables and progresses through data inspection, statistical analysis, data quality checking, exploratory analysis, preprocessing, classification, validation, tuning, prediction, and interpretation.

The analytics component provides an understanding of customer characteristics and churn patterns through descriptive statistics and visualizations. The machine learning component transforms these insights into predictive models using Logistic Regression, Decision Tree, Random Forest, and Gradient Boosting. Model evaluation is performed using multiple classification metrics, while stratified cross-validation and hyperparameter tuning provide additional validation.

The final tuned Random Forest produces test accuracy of 0.9418, precision of 0.9768, recall of 0.8662, F1-score of 0.9182, and ROC-AUC of 0.9317. Feature importance analysis further provides a model-specific interpretation of the variables contributing to prediction.

Overall, the project provides practical experience in transforming raw customer data into meaningful analytical information and machine learning predictions. It demonstrates that a complete data science workflow requires more than model training: data understanding, quality checking, preprocessing, exploratory analysis, validation, evaluation, and interpretation are all important components of a reliable project.

The project can serve as a foundation for future work involving deployment of the prediction model, automated dashboards, probability-based customer segmentation, monitoring of model performance, and integration with real-time customer data when appropriate.

REFERENCES

The references section records the primary technical resources and project materials used for implementation and understanding of the workflow.

- Project dataset: Customer Churn Prediction Excel dataset, raw_data sheet.
- Python documentation and standard Python programming resources.
- Pandas documentation for data loading, manipulation, and descriptive analysis.
- NumPy documentation for numerical computation.
- Matplotlib documentation for visualization.
- Seaborn documentation for statistical visualization.
- Scikit-learn documentation for preprocessing, classification, model evaluation, cross-validation, and GridSearchCV.
- Google Colab notebook containing the complete project implementation and outputs.
- Customer Churn Analytics and Machine Learning project notebook/PDF generated during project development.

The analytical results and numerical values presented in this report are based on the completed project notebook and its generated outputs. The report follows the structure of the supplied academic reference while expanding each major section into a dedicated, detailed page.